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UNAUTHORIZED MEMORY ACCESS MITIGATION

Abstract

Apparatuses and methods related to mitigating unauthorized memory access are described. Mitigating unauthorized memory access can include verifying whether an access command is authorized to access a protected region of a memory array. The authorization can be verified utilizing a key and a memory address corresponding to the access command. If an access command is authorized to access a protected region, then a row of the memory array corresponding to the access command can be activated. If an access command is not authorized to access the protected region, then a row of the memory array corresponding to the access command may not be activated.

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Background/Summary

PRIORITY INFORMATION [0001] This application is a Continuation of U.S. application Ser. No. 18/464,015, filed Sep. 8, 2023, which issues as U.S. Pat. No. 12,321,617, on Jun. 3, 2025, which is a Continuation of U.S. application Ser. No. 17/665,307, filed Feb. 4, 2022, which issued as U.S. Pat. No. 11,755,210 on Sep. 12, 2023, which is a Continuation of U.S. application Ser. No. 16/235,303, filed Dec. 28, 2018, which issued as U.S. Pat. No. 11,256,427 on Feb. 22, 2022, the specification of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to memory, and more particularly to apparatuses and methods associated with protecting memory using a key.

BACKGROUND

[0003] Memory devices are typically provided as internal, semiconductor, integrated circuits in computers or other electronic devices. There are many different types of memory including volatile and non-volatile memory. Volatile memory can require power to maintain its data and includes random-access memory (RAM), dynamic random access memory (DRAM), and synchronous dynamic random access memory (SDRAM), among others. Non-volatile memory can provide persistent data by retaining stored data when not powered and can include NAND flash memory, NOR flash memory, read only memory (ROM), Electrically Erasable Programmable ROM (EEPROM), Erasable Programmable ROM (EPROM), and resistance variable memory such as phase change random access memory (PCRAM), resistive random access memory (RRAM), and magnetoresistive random access memory (MRAM), among others.

[0004] Memory is also utilized as volatile and non-volatile data storage for a wide range of electronic applications. Non-volatile memory may be used in, for example, personal computers, portable memory sticks, digital cameras, cellular telephones, portable music players such as MP3 players, movie players, and other electronic devices. Memory cells can be arranged into arrays, with the arrays being used in memory devices.

[0005] Various computing systems include a number of processing resources that are coupled to memory (e.g., a memory system), which is accessed in association with executing a set of instructions (e.g., a program, applications, etc.). For various reasons, it can be desirable to prevent unauthorized access to memory (e.g., via read and/or write operations) or particular portions thereof. For instance, a memory system may store sensitive data (e.g., data desired to be kept secret, such as passwords, personal information, etc.).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a block diagram of an apparatus in the form of a computing system including a memory device in accordance with a number of embodiments of the present disclosure.

[0007] FIG. 2 is a block diagram of an apparatus in the form of a memory device including a memory array and portions of a controller capable of protecting regions of memory using a key in

accordance with a number of embodiments of the present disclosure.

[0008] FIG. 3 is a block diagram of a region of a memory array in accordance with a number of embodiments of the present disclosure.

[0009] FIG. 4 illustrates an example flow diagram of a method for processing a security mode initialization command in accordance with a number of embodiments of the present disclosure.

[0010] FIG. 5 illustrates an example flow diagram of a method for accessing a protected region of a memory array in accordance with a number of embodiments of the present disclosure.

[0011] FIG. 6 illustrates an example flow diagram of a method for accessing a protected region of a memory array in accordance with a number of embodiments of the present disclosure.

[0012] FIG. 7 illustrates an example machine of a computer system within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed.

DETAILED DESCRIPTION

[0013] The present disclosure includes apparatuses and methods related to mitigating unauthorized memory access. Access commands can be provided from a host to a memory device. The memory device may rely on the host for implementing security measures to prevent unauthorized access to the memory device. However, implementing security measures at the memory device may further improve security and may mitigate unauthorized memory accesses.

[0014] In prior approaches the host can manage keys used to access a memory device. That is, in prior approaches a memory device does not manage keys used to access protected regions of a memory array of the memory device.

[0015] In various embodiments, a memory device can mitigate unauthorized memory accesses by verifying access commands as authorized utilizing credentials provided along with, or as part of, the access commands. The credentials can be stored in a plurality of registers implemented in the memory device prior to receipt of the access command (e.g., from a host). As used herein, an access command can include one or more sub-commands. For example, an access command can include a pre-charge command, an activate command, a read command, and/or a write command, among other possible commands.

[0016] The authorization of an access command can be verified utilizing a key (e.g., credential(s)). The access command can request access to an address and/or a plurality of addresses. The memory device can determine whether the address is locked or unlocked based on a security mode associated with the address. If the address is locked, then the memory device can refrain from providing access to the address unless a key associated with the access command is also provided to the memory device. The key can be verified against a stored key to determine whether to unlock the address.

[0017] If the key matches the stored key, then the memory device can unlock the address and can provide access to the address(es). If the key does not match the stored key, then the memory device can refrain from providing access to the address(es).

[0018] Implementing security measures at a memory device to prevent unauthorized access can increase the security of the memory device beyond the security which may be provided by a host. For example, unauthorized access commands may be prevented from accessing the memory device by security measures implemented at a host as well as by security measures implemented at a memory device.

[0019] As used herein, “a number of” something can refer to one or more of such things. For example, a number of memory devices can refer to one or more memory devices. A “plurality” of something intends two or more. Additionally, designators such as “N,” as used herein, particularly with respect to reference numerals in the drawings, indicates that a number of the particular feature so designated can be included with a number of embodiments of the present disclosure.

[0020] The figures herein follow a numbering convention in which the first digit or digits correspond to the drawing figure number and the remaining digits identify an element or

component in the drawing. Similar elements or components between different figures may be identified by the use of similar digits. As will be appreciated, elements shown in the various embodiments herein can be added, exchanged, and/or eliminated so as to provide a number of additional embodiments of the present disclosure. In addition, the proportion and the relative scale of the elements provided in the figures are intended to illustrate various embodiments of the present disclosure and are not to be used in a limiting sense.

[0021] FIG. 1 is a block diagram of an apparatus in the form of a computing system **100** including a memory device **120** in accordance with a number of embodiments of the present disclosure. As used herein, a memory device **120**, a memory array **130**, and/or host **110** might also be separately considered an “apparatus.”

[0022] In this example, system **100** includes a host **110** coupled to memory device **120** via an interface **156**. The computing system **100** can be a personal laptop computer, a desktop computer, a digital camera, a mobile telephone, a memory card reader, or an Internet-of-Things (IoT) enabled device, among various other types of systems. Host **110** can include a number of processing resources (e.g., one or more processors, microprocessors, or some other type of controlling circuitry) capable of accessing memory **120**. The system **100** can include separate integrated circuits, or both the host **110** and the memory device **120** can be on the same integrated circuit. For example, the host **110** may be a system controller of a memory system comprising multiple memory devices **120**, with the system controller **110** providing access to the respective memory devices **120** by another processing resource such as a central processing unit (CPU).

[0023] In the example shown in FIG. 1, the host **110** is responsible for executing an operating system (OS) **103** and/or various applications that can be loaded thereto (e.g., from memory device **120** via controller **140**). The host **110** may not be responsible for managing keys or addresses that define the protected region. The host **110** can provide access commands and/or security mode initialization commands to a memory device via the interface **156**. In some examples, the memory device may provide an indication of an unauthorized access attempt. In various embodiments, the unauthorized access attempt can result in providing data to the host such that the issuer of the unauthorized access attempt is not notified of the memory device's identification of the unauthorized access attempt. The data received from the memory device **120** may not be the data stored by the memory cells having the address provided by the access command. The host **110** may receive data generated responsive to identifying the access command as unauthorized.

[0024] For clarity, the system **100** has been simplified to focus on features with particular relevance to the present disclosure. The memory array **130** can be a DRAM array, SRAM array, STT RAM array, PCRAM array, TRAM array, RRAM array, NAND flash array, and/or NOR flash array, for instance. The array **130** can comprise memory cells arranged in rows coupled by access lines (which may be referred to herein as word lines or select lines) and columns coupled by sense lines (which may be referred to herein as digit lines or data lines). Although a single array **130** is shown in FIG. 1, embodiments are not so limited. For instance, memory device **120** may include a number of arrays **130** (e.g., a number of banks of DRAM cells).

[0025] The memory device **120** includes address circuitry **142** to latch address signals provided over an interface **156**. The interface can include, for example, a physical interface employing a suitable protocol (e.g., a data bus, an address bus, and a command bus, or a combined data/address/command bus). Such protocol may be custom or proprietary, or the interface **156** may employ a standardized protocol, such as Peripheral Component Interconnect Express (PCIe), Gen-Z, CCIX, or the like. Address signals are received and decoded by a row decoder **146** and a column decoder **152** to access the memory array **130**. Data can be read from memory array **130** by sensing voltage and/or current changes on the sense lines using sensing circuitry **150**. The sensing circuitry **150** can comprise, for example, sense amplifiers that can read and latch a page (e.g., row) of data from the memory array **130**. The I/O circuitry **144** can be used for bi-directional data communication with host **110** over the interface **156**. The read/write circuitry **148** is used to write

data to the memory array **130** or read data from the memory array **130**. As an example, the circuitry **148** can comprise various drivers, latch circuitry, etc.

[0026] Controller **140** decodes signals provided by the host **110**. These signals can include chip enable signals, write enable signals, and address latch signals that are used to control operations performed on the memory array **130**, including data read, data write, and data erase operations. In various embodiments, the controller **140** is responsible for executing instructions from the host **110**. The controller **140** can comprise a state machine, a sequencer, and/or some other type of control circuitry, which may be implemented in the form of hardware, firmware, or software, or any combination of the three.

[0027] In accordance with various embodiments, the controller **140** can be configured to decode a security mode initialization command received thereto. The security mode initialization command can be received from the host **110**. The security mode initialization command can be provided to the memory device **120** to set a security mode of the memory device **120** and/or to designate one or more protected regions of the memory device **120**. A security mode can include a locked mode and an unlocked mode. The memory device **120** can be configured to provide access to a protected region of the memory array **130** if the memory device **120** is in an unlocked mode or to prevent access to the protected region of the memory array **130** if the memory device **120** is in a locked mode.

[0028] The OS **103**, as executed by the host **110**, can initialize the security mode initialization command to store a key and an address or a range of addresses of the memory array **130** in one or more registers of the controller **140**. The stored key and address can define the protected region of the memory array **130**. The OS **103** can initialize the security mode initialization command during an initialization of the OS **103** or a time after the OS **103** is initialized.

[0029] The address received from the host **110** can be a logical address. A logical address can be translated, mapped, or resolved to a physical address. A logical address can include a virtual address, for example. The physical address may be used by the memory device **130** to access data from a memory array **130**. The physical address can be directly mapped to a memory cells or row of the memory array **130**. The logical address corresponding to the initialization command and/or access commands and received from the host **110** can be translated by the memory device **120** to generate the physical address. The physical address can be a physical row address of the memory array **130**.

[0030] The key can be a security token used to gain access to a protected region of the memory array **130**. The key can be encrypted or unencrypted. The key can be provided by the OS **103** and used by the OS **103** to access the protected region of the memory array **130**. The key can be unique to a protected region of memory and/or can be associated with a plurality of protected regions of memory. As described further below, the key can comprise one or more bits which can be stored in one or more registers of the memory device **120**.

[0031] The protected region of the memory array **130** describes a region of the memory array **130** that is protected using the key. The protected range can be defined by a first memory address and a second memory address. The first memory address can be a starting address and the second memory address can be an ending address. In some examples, the protected range is stored as a starting address and as an offset. The offset together with the starting address can be used to generate the ending address. The protected region can be continuous from the starting address to the ending address.

[0032] In some examples, the memory array **130** can comprise one or more protected regions. Each of the protected regions can be defined using a starting address and an offset. Each of the starting addresses corresponding to a different protected region can be unique and/or can be a same starting address. Each of the offsets can also be a same offset or a different offset.

[0033] In various instances, the host **110** can provide an access command to the memory device **120**. The access command can be provided to access a protected region of the memory device **120**.

The access command can be associated with an address or a range of addresses and a key. The memory device **120** can compare the provided address to a protected range to determine whether the address is within the protected range. If the address is within the protected range, the memory device **120** can compare the key with a stored key to determine whether the key and the stored key match. If the key matches the stored key, then the memory device can enter an unlocked mode from a locked mode. The memory device **120** can, via the controller **140**, enable a row driver to activate a row of the memory array **130** corresponding to the address (e.g., protected region). If the key does not match, the memory device **120** can, via the controller **140**, prevent access to the protected region by preventing enablement of the row driver **147** of the memory array **130**, thus preventing activation of a row corresponding to the access command.

[0034] FIG. **2** is a block diagram of an apparatus in the form of a memory device **220** including a memory array **230** and portions of a controller capable of protecting regions of memory using a key in accordance with a number of embodiments of the present disclosure. The memory device **220** is analogous to memory device **120** in FIG. **1**. The memory device **220** includes the memory array **230** and portion of the controller **140** in FIG. **1**.

[0035] The controller can include a command decoder **221**, mode registers **224**, a key register **226**, protected region registers **228**, and an access counter register **231**. The controller can also include the address match unit **222** and a key match unit **223**.

[0036] In this example, the interface (e.g., **156** shown in FIG. **1**) comprises an address bus **256-1**, a command bus **256-2**, and a data bus **256-3**. The device **220** can receive the security mode initialization command and/or access commands along with keys via the command bus **256-2**. The device **220** can receive addresses via the address bus **256-1**, and data can be provided to/from the device **220** via the data bus **256-3**.

[0037] A host can provide, via the command bus **256-2**, the security mode initialization command to initialize a security mode of the memory device **220**. The memory device **220** can receive the security mode initialization command at the command decoder **221**. The command decoder **220** can decode the security mode initialization command.

[0038] In some examples, the security mode initialization command can be associated with a key and a number of addresses received via the command bus **256-2** and the address bus **256-1**.

[0039] The controller can store a key in the key register **226** and can store the one or more addresses in the protected region registers **228**. Each of the mode registers **224**, the key registers **226**, the protected region registers **228**, and/or the access counter register **231** can be comprised of one or more registers.

[0040] The one or more addresses can be stored in the protected region registers **228** as a starting address and an offset. The starting address can be stored in a first register of the protected regions registers **228** and the offset can be stored in a second register of the protected region registers **228**. The starting address and the ending address can define a region of the memory array **230**. As such, the region to protect (e.g., protected region) can be stored in the protected region register **228**.

[0041] The key can be stored in the key register **226**. In some examples, a plurality of keys can be stored in one or more key registers including the key register **226**. Each of the plurality of keys can be associated with a different one of the plurality of protected regions stored in the protected region registers including the protected region register **228**. The plurality of keys can be used to allow access to the protected regions. For example, a first key can be used to allow access to a first protected region and a second key can be used to allow access to a second protected region.

[0042] Responsive to storing the key in the key register **226** and the address in the protected region register **228**, the controller can change a security mode of the memory device **220** from an unlocked mode to a locked mode in the mode registers **224**. The mode registers **224** can include a security mode register. The security mode register can store a first value representing an unlocked mode or a second value representing a locked mode, among other possible modes. The locked mode can be used to prevent access to a protected region of the memory array **230**. An unlocked

mode can be used to allow access to a protected region of the memory array **230**.

[0043] In some examples, responsive to receipt of the security mode initialization command, the controller can set an access counter register **231**. For example, the access counter register **231** can be set to zero. The access counter register **231** can provide a count of access commands directed to the protected region of the memory array **230** (e.g., as defined by protected region register **228**).

[0044] The controller can also process access commands. For example, an access command received via the command bus **256-2** can be decoded by the command decoder **221**. The address match unit **222** can receive an address corresponding to the access command at the address match unit **222** of the controller. The address match unit **222** can determine whether the received address is within a protected region (e.g., as stored in the protected region register **228**).

[0045] If the received address is in a protected region, then the controller, via the key match unit **223**, can determine whether the key associated with the access command matches a key stored in the key register **226**. If the key associated with the access command matches the key stored in the key register **226**, then the controller can modify the mode registers **224** from a locked mode to an unlocked mode.

[0046] The controller can provide a signal to the row drivers **247** to activate a row corresponding to the received address if the mode registers **224** reflect an unlocked mode. The controller can prevent a signal from being provided to the row drivers **247** if the mode registers **224** reflect a locked mode. Although the row drivers **247** are shown as being in the memory array **230**, the row drivers **247** can also be implemented externally to the memory array **230** as shown in FIG. **1**.

[0047] The controller can also include the access counter register **231**. The access counter register can store a count corresponding to a quantity of access commands received at the memory device **220**, where the access commands are associated with an address in the protected region. Prior to allowing access, by the access commands, to the array **230**, the controller can determine whether the access count stored in the access counter register **231** is greater than a threshold. If the access count is greater than the threshold, then the controller can refrain from allowing access to the address corresponding to an access command. If the access count is less than the threshold, then the controller may allow access to the address provided that the address is in a protected region and that the key corresponding to the access command matches a key stored in the protected region register **228**.

[0048] The controller can determine whether the access count is greater than a threshold before it is determined whether the address is in the protected region, before it is determined whether the key associated with the access command matches the key stored in the key register **226**, and/or after it is determined whether the key associated with the access command matches the key stored in the key register **226**.

[0049] The controller can determine whether the access count is greater than the threshold to protect against row hammer attacks, for example. Row hammer attacks access memory cells of the memory array **230** to leak the charge stored in the memory cells among themselves to change the content of nearby memory cells. By preventing access to memory cells above a threshold, the controller can protect against row hammer attacks. In some examples, the controller can report any unauthorized access attempt to the protected region. The threshold can be used to determine when to report each of the unauthorized access attempts logged before the threshold is reached.

[0050] FIG. **3** is a block diagram of a region of a memory array **330** in accordance with a number of embodiments of the present disclosure. The protected region of the memory array can comprise protected rows **345** and/or padding rows **343-1** and **343-2**.

[0051] In some examples, the controller can add the padding rows **343-1** and/or **343-2** to the protected rows **345**. The protected row can correspond to a protected region defined by the security mode initialization command. The memory addresses associated with the security mode initialization command can correspond to the protected rows **345**. The controller can pad the protected region by adding the padding rows **343-1** and **343-2** to the protected rows **345** to generate

the padded protected region.

[0052] The padding rows **343-1** and **343-2** can be added to the protected rows **345** to protect against row hammer attacks. The padding rows **343-1-** and **343-2** can be utilized in conjunction with the access counter of the padded protection region to protect the protected rows **345** from unauthorized memory access utilizing row hammer attacks, for example.

[0053] FIG. **4** illustrates an example flow diagram of a method for processing a security mode initialization command in accordance with a number of embodiments of the present disclosure. At **435**, an OS can initiate a security mode initialization command. The security mode initialization command can be provided by the OS to define a protected region of a memory array and to associate a key with the protected region.

[0054] Defining a protected region utilizing the security mode initialization command provides the OS flexibility. The OS can have flexibility to define the size and content of a protected region of the memory array. For example, the OS can define the protected region as comprising a first size or a second size, among a number of other sizes. The OS can activate a security mode by providing the security mode initialization command or can refrain from activating the security mode by refraining from providing the security mode initialization command to the memory device.

[0055] A memory device can function in a security mode or without the security mode based on the OS's selection. Further, the OS can define a size or location of the protected region after the protected region has been initialized. For example, after initialization of a security mode, the OS can increase the size of the protected region or decrease the size of the protected region. After initialization of a security mode, the OS can also change a base address of the protected region and/or an offset of the protected region. The OS can also exit the security mode after the memory device has been placed in the security mode. For example, the OS can store a predefined value in the protected region registers. The OS can store a zero as a base address and/or an offset of the protected region to exit the security mode.

[0056] The OS can utilize an application programming interface (API) to generate the security mode initialization command and/or an access command used to define and/or access a protected region of the memory array. The OS can comply with the security features of the memory device utilizing the API.

[0057] In some examples, the security mode initialization command can be generated by the OS and/or received by the memory device as part of an initialization process of a computing device and/or the OS. At **437**, the memory device can store a key in a key register responsive to receiving the security mode initialization command. At **439**, the memory device can store an address of a memory device in a protected region register. The memory device can set a mode register based on the storing of the key and/or the storing of the address. The mode register can identify whether region of the memory array is protected. The memory device can set the mode register to a locked mode. The locked mode can be a protected mode.

[0058] FIG. **5** illustrates an example flow diagram of a method for accessing a protected region of a memory array in accordance with a number of embodiments of the present disclosure. At **551**, the authorized process that has the key gains access to the security region. The process can be an instance of a program that is being executed by the host such as an application process. For example, the process can be an OS and/or a different application, among other possible processes. The process can be authorized upon verifying that a key associated with an access command provided by the process matches a stored key in the key register.

[0059] At **553**, the enablement bit flag can be flipped. The enablement bit flag can be stored in a security mode register shown in FIG. **2** as one of the mode registers **224**. The enablement bit flag can be flipped from a locked mode to an unlocked mode. At **555**, the authorized process can read from the secured memory region. An access command can be a read command or a write command, among other possible access commands. The controller can prevent enabling a row driver from activating a row responsive to the enablement bit flag indicating the locked mode,

where the row corresponds to a received address associated with the access command. For example, the controller can prevent any row driver enablement when the mode bit indicated that a secured memory region is locked.

[0060] At **557**, the authorized process can conclude reading from the protected region. At **559**, the controller can return the enablement bit flag to its original value. For example, the enablement bit flag can be returned to a locked mode.

[0061] FIG. **6** illustrates an example flow diagram of a method for accessing a protected region of a memory array in accordance with a number of embodiments of the present disclosure. The method can be executed by a memory device of a computing system.

[0062] At **660**, a memory device can receive an access command corresponding to an address of a memory array to access. At **662**, a determination can be made as to whether the address is within a protected region of the memory arrays as indicated by a protected region register. The address can be within the protected region if the address is greater than a starting address of the protected region but less than the ending address of the protected region. In some examples, the access command can be associated with a plurality of addresses to access. The plurality of addresses can be within the protected region if at least one of the plurality of addresses is greater than a starting address and at least one of the of the plurality of addresses is less than the ending address of the protected region.

[0063] At **664**, responsive to determining that the address is within the protected region, the controller can determine whether a first key associated with the access command matches a second key stored in a key register of the memory device. The first key can match the second key if the first key is equal to the second key or if the first key is substantially equal to the second key. The first key can match the second key if the second key can be derived from the first key, among other examples.

[0064] At **668**, responsive to determining that the first key matches the second key, the controller can allow access to the address of the memory array. The memory device can allow access to the address by providing the data stored at memory cells having the address to the host. Access can be allowed by allowing an activation of a row of the array corresponding to the address. The method can further allow activation of a row to which the address corresponds by switching an access enable flag from a first state to a second state, wherein rows corresponding to addresses within the protected region are not activatable unless the access enable flag is in the second state. Subsequent to execution of the access command (e.g., read command), the access enable flag can be returned to the first state to prevent activation of rows corresponding to the protected region.

[0065] At **670**, responsive to determining that the first key does not match the second key, the controller can prevent access to the address of the memory array. Responsive to determining that the address is outside the protected region, allowing access to the address without performing a key match comparison.

[0066] In some examples, the access command can be a read command. Responsive to determining that the address is within the protected region and that the first key does not match the second key, the method may further comprise returning data read from a row corresponding to a different address of the array. In some embodiments, the first key and the address can be provided in a same cycle as a cycle in which the access command is received. In other embodiments, the first key and the address can be provided in a different cycle as a cycle in which the access command is received.

[0067] FIG. **7** illustrates an example machine of a computer system **740** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed. In some embodiments, the computer system **740** can correspond to a host system (e.g., the system **110** of FIG. **1**) that includes, is coupled to, or utilizes a memory sub-system (e.g., the memory device **120** of FIG. **1**) or can be used to perform the operations of a controller (e.g., the controller **140** of FIG. **1**, including the register **224**, **226**, **228**, and **231** of FIG.

2). In alternative embodiments, the machine can be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine can operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

[0068] The machine can be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

[0069] The example computer system **740** includes a processing device **702**, a main memory **704** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **706** (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage system **718**, which communicate with each other via a bus **730**.

[0070] Processing device **702** represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device can be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **702** can also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **702** is configured to execute instructions **726** for performing the operations and steps discussed herein. The computer system **740** can further include a network interface device **708** to communicate over the network **720**.

[0071] The data storage system **718** can include a machine-readable storage medium **724** (also known as a computer-readable medium) on which is stored one or more sets of instructions **726** or software embodying any one or more of the methodologies or functions described herein. The instructions **726** can also reside, completely or at least partially, within the main memory **704** and/or within the processing device **702** during execution thereof by the computer system **740**, the main memory **704** and the processing device **702** also constituting machine-readable storage media.

[0072] In one embodiment, the instructions **726** include instructions to implement functionality corresponding to the controller **140** of FIG. 1. While the machine-readable storage medium **724** is shown in an example embodiment to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

[0073] Although specific embodiments have been illustrated and described herein, those of ordinary skill in the art will appreciate that an arrangement calculated to achieve the same results can be substituted for the specific embodiments shown. This disclosure is intended to cover adaptations or variations of various embodiments of the present disclosure. It is to be understood that the above description has been made in an illustrative fashion, and not a restrictive one. Combinations of the above embodiments, and other embodiments not specifically described herein will be apparent to those of skill in the art upon reviewing the above description. The scope of the

various embodiments of the present disclosure includes other applications in which the above structures and methods are used. Therefore, the scope of various embodiments of the present disclosure should be determined with reference to the appended claims, along with the full range of equivalents to which such claims are entitled.

[0074] In the foregoing Detailed Description, various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the disclosed embodiments of the present disclosure have to use more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

Claims

1. An apparatus, comprising: a memory array; a key register configured to store a first key used to determine whether access commands are allowed access to a protected region of the memory array; a protected region register configured to store addresses defining the protected region; an access count register to store an access count; and control circuitry configured to: responsive to receiving a security mode initialization command from a host: store the first key in the key register; store a first address corresponding to the memory array in the protected region register; store a third address corresponding to the memory array in the protected region register, wherein the first address is a protected address, the third address is a padding row and the first key corresponds to the protected address and the padding row; determine whether a second address corresponding to an access command is within the protected region or the padding row and determine whether a second key from a number of keys corresponding to the access command from a number of access commands matches the first key stored in the key register; responsive to determining that the second address is within the protected region or the padding row and that the second key matches the first key, allow access to the second address based on a determination the access count is less than a threshold; and responsive to allowing access to the second address, increment the access count register.
2. The apparatus of claim 1, wherein the padding row is adjacent to a beginning address of the protected region.
3. The apparatus of claim 1, wherein the padding row is adjacent to a last address of the protected region.
4. The apparatus of claim 1, wherein the controller is further configured to add the padding row to the protected region.
5. The apparatus of claim 1, wherein the controller comprises access count register configured to store the access count.
6. The apparatus of claim 5, wherein the controller is configured to, responsive to receiving the security mode initialization command, setting the access count registers to a predefined value.
7. The apparatus of claim 5, wherein the controller is further configured to, responsive to receiving the security mode initialization command, setting the access count registers to a 0-value such that the access count is set to a 0-value.
8. The apparatus of claim 7, wherein the access count provides a count of access commands directed to the protected region.
9. The apparatus of claim 1, wherein the access count provides a count of access commands to the padding rows.
10. A method comprising: adding padding rows to a protected region to generate a padded protected region; receiving, at a memory device, an access command corresponding to an address of a memory array to access; determining whether the address is within the padded protected region

of the memory array as indicated by a protected region register; responsive to determining that the address is within the padded protected region, incrementing an access count stored in an access count register wherein the access count includes a quantity of access commands to the address in the padded protected region received at the memory device; responsive to determining that the address is within the protected region, determining whether a first key, from a number of keys, associated with the access command, from a number of access commands, matches a second key stored in a key register of the memory device; responsive to determining that the first key matches the second key, determining that the access count is less than a threshold; and responsive to determining that the access count is less than the threshold, allowing access to the address of the memory array.

11. The method of claim 10, wherein the padding rows include first rows adjacent to starting address of the protected region and second rows adjacent to an ending address of the protected region.

12. The method of claim 10, further comprising receiving the access command, the address, and the first key from a host to which the memory device is coupled via an interface.

13. The method of claim 10, wherein allowing access to the address of the memory array further comprises allowing an activation of a row of the array corresponding to the address.

14. The method of claim 10, further comprising, responsive to determining that the address is outside the protected region, allowing access to the address without performing a key match comparison.

15. The method of claim 10, wherein the access command is a read command, and wherein allowing access to the address of the memory array comprises: allowing activation of a row to which the address corresponds by switching an access enable flag from a first state to a second state, wherein rows corresponding to addresses within the protected region are not activatable unless the access enable flag is in the second state; and subsequent to execution of the read command, returning the access enable flag to the first state to prevent activation of rows corresponding to the protected region.

16. An apparatus, comprising: a memory array; a mode register configured to identify whether a received address that is within a protected region or padding rows is protected; a protected region register configured to store addresses defining the protected region of the memory array and the padding rows of the memory array; an access count register to store an access count of the protected region and the padding rows; a key register configured to store a second key used to determine whether access commands are allowed access to the protected region and the padding rows; and control circuitry configured to: determine whether the access count is greater than a threshold; responsive to determining that the access count is not greater than the threshold, determine whether the received address is in the protected region or the padding rows of the memory array; access the mode register to determine whether a first key matches the second key stored in the key register, wherein the first key is provided with the access command; responsive to determining that the received address corresponds to the protected region and responsive to determining that the first key matches the second key, enable a row driver of to activate a row, of the memory array, corresponding to the received address and increment the access count.

17. The apparatus of claim 16, wherein the control circuitry is further configured to determine whether the received address is greater than a starting address of the protection region or the padding rows and less than an ending address of the protected region or the padding rows.

18. The apparatus of claim 16, the control circuitry is configured to activate the row by providing a signal to the row driver of the memory array.

19. The apparatus of claim 16, wherein the control circuitry is further configured to: receive an access command; responsive to receiving the access command, determine whether the first key corresponding to the access command matches the second key; and responsive to determining that the first key matches the second key, store a value representing a matching between the first key

and the second key in the mode register.

20. The apparatus of claim 19, wherein the control circuitry is further configured to, responsive to determining that the first key does not match the second key, store a different value, representing that there is no match, in the mode register.
